

Microeconomic Foundations, I: Choice and Competitive Markets

Instructor's Manual

Chapter 11: Classic Demand Theory

Notes on the Chapter

This chapter completes classic demand theory, deriving and interpreting Roy's Identity and the Slutsky Equation, providing a duality analysis of the indirect utility function, giving sufficient conditions for Marshallian demand to be differentiable (the usual bordered Hessian conditions), and then discussing integrability.

Please note that the chapter *discusses* integrability. I debated (myself and colleagues) long and hard about whether to give a complete and rigorous treatment of the Integrability Theorem and, in the end, decided not to do so. My reason is that I don't believe the required excursion into the theory of partial differential equations is worth the effort. But it was a close call, and I appreciate that some instructors will be upset that, of all places in the book to engage in hand-waving, I chose this one.

Perhaps worse still from the perspective of the classic theory of the consumer, I do not fully connect revealed preference approaches to the sufficient conditions in the Integrability Theorem, except for the easy bits about Walras' Law, homogeneity of demand, and negative semi-definiteness of the Slutsky matrix (assuming differentiability); the classic developments of Samuelson, Houthakker, as well as more modern extensions and elaborations, are not tackled. At least one colleague has excoriated me about this choice, and I apologize to him and to anyone else offended by this omission.

I've always found this chapter to be a grind to teach. This probably reflects in part my "tin ear" when it comes to this subject. And it may reflect exhaustion (mine and the students') after tackling the heavy bits of Chapters 9 and 10. The details of the proof of the differentiability of indirect utility are painful and, after detailed in-class exposition of duality applied to the profit function/production possibilities set in Chapter 9 and to the expenditure function/utility function in Chapter 10, doing "it" once more for indirect utility doesn't add a lot. I do spend time in class explaining how the proof of differentiability of Marshallian demand goes and, in particular, why differentiability won't typically survive a change in the basis (the set of binding constraints) in the consumer's problem.

If anyone can suggest to me how to make it more interesting/exciting/fun, perhaps I'll be able to modify these dour notes.

Notes on the Problems

If you want to give students relatively easy practice with the Implicit Function Theorem, Problem 3 is good. If you are pushing the students to go very deeply into precisely what has and has not been shown in the text, Problem 5 is a good "detective problem." Problem 7 has the virtue of making the abstract results of the chapter at least somewhat concrete. Note that the "solution" I provide below to Problem 2 only sketches the required arguments; this problem is a real mess to do exactly, so if you assign this problem, you might want to be kind to your students and tell them that you'll accept sketches. (I have half a suspicion that the last part—differentiability of e in (p, v) —can be gotten by a clever application of the Implicit Function Theorem. But, if so, I'm insufficiently clever. If someone could help me out with this, I'd be grateful.)

Solutions to Non-Starred Problems

■ 11.2. Extending Proposition 11.1 to differentiability at points $(p^0, 0)$ faces the difficulty that, at $y = 0$, no good is being consumed at strictly positive levels. So we need a different way of identifying the "right" commodities. Fixing p^0 , let b_i be the bang-for-the-buck of good i at these prices and at the consumption bundle $x = 0$; that is $b_i = (1/p_i^0)(\partial u / \partial x_i |_{x=0})$. Let $b^* = \max\{b_i; i = 1, \dots, k\}$. Then under the conditions of Proposition 11.1 (the consumer is locally insatiable, u is continuously differentiable, and the solution to the consumer's problem is unique in a neighborhood of $(p^0, 0)$), the indirect utility function ν is continuously differentiable at the point $(p^0, 0)$ in (p, y) , with partial derivatives equal to 0 in all k prices and $\partial \nu / \partial y = b^*$. (In fact, we don't really need to assume that the solution is unique in a neighborhood of $(p^0, 0)$; we know it is unique at that point, of course, and that suffices. And if u is not locally insatiable at 0, then ν will be the constant $u(0)$ over some neighborhood of $(p^0, 0)$, hence continuously differentiable with gradient vector 0.)

I will not supply a detailed proof; it is (at best) a mess of epsilons that hides a very simple idea: The question is, essentially, how much incremental utility (over the utility of the bundle of all zeros) can you purchase with a small amount of money, when prices are close to (and approaching) a strictly positive price vector p^0 . A lower bound can be established by spending all your money on any one of the goods that has the maximal bang-for-the-buck; that is, on any j such that $b_j = b^*$. If you have y to spend, this will give you a utility increment (to a first-order term) of b^*y . And you can't do any better than that: However you divide your income y amongst the k goods, if we let y_i be your expenditure on good i , then your increment in utility is, to a first-order term, $\sum_i b_i y_i$. Making this precise is something of a mess because the prices are moving, but as they are moving in vanishingly small neighborhoods of a strictly positive price vector p^0 , that movement is inconsequential. And note that a small change in prices doesn't affect your "max utility" at all, since at $y = 0$ you have nothing to spend.

The second part of this problem asks, What happens when u is not continuously differentiable? (Of course, we'll assume that u is continuous and locally insatiable.) Recall in this regard that having a nondifferentiable u had no impact on our earlier results (back in Chapter 10) on the differentiability of e in p . But nondifferentiability of u affects the differentiability of ν in both p and y . To see this, consider

$$u(x) = \begin{cases} x, & \text{if } x \leq 1 \text{ and} \\ 2x - 1, & \text{if } x > 1. \end{cases}$$

(We're just putting a kink in the utility function at the value $x = 1$.) In this one-dimensional world, it is clear that the consumer's problem always has a unique solution; namely, spend all your money on the one good. Hence

$$\nu(p, y) = \begin{cases} y/p, & \text{if } y/p \leq 1 \text{ and,} \\ 2(y/p) - 1, & \text{if } y/p > 1. \end{cases}$$

The kink reappears in both p and y , whenever we are at the "critical" values $p = y$. The intuition here should be clear: In the EMP, once we fix v , the feasible set is fixed, and the "local behavior" of u is inconsequential. We can vary p however we want, and still the local behavior of u doesn't matter. But in the consumer's problem, shifting p shifts the feasible set, and so how u behaves along the border of the feasible set does matter.

Although you weren't asked about this in the problem, note that this intuition also speaks to the assumption in Proposition 11.1 that u is locally insatiable, something else that was not assumed in the corresponding proposition (Proposition 10.6) concerning differentiability of e . If u is locally satiated at some point x^0 , then for prices p^0 and income y^0 such that x^0 is the unique solution of the CP at (p^0, y^0) , ν will be locally constant, hence have a gradient vector identically equal to zero. "Local insatiability" in the EMP holds for any continuous utility function u , in the sense that the objective function $p \cdot x$ is strictly increasing locally at every point $x^0 \neq 0$.

Finally, the problem asks about differentiability of e in v or, more precisely, jointly in (p, v) . I again spare you the gruesome details, but here's the result and broad sketch of how the proof goes: We will have differentiability if (a) u is continuously differentiable and, moreover, if (b) $\partial u / \partial x_i > 0$ for some commodity i , at every point x^0 . (I'm not saying that condition b is necessary, but the proof of differentiability of e that I'm about to sketch needs this.) The proof runs as follows: As v changes, the feasible set changes, so as in the proof of Proposition 11.1 (and unlike in the proof of Proposition 10.6, which was actually the proof of Proposition 9.22), as v moves, you will need to "correct" the solutions. Fixing some (p^0, v^0) and a sequence $\{(p^n, v^n)\}$ with (p^0, v^0) as its limit, let x^0 be the (assumed unique) solution of the EMP at (p^0, v^0) and let x^n be some solution of the EMP at (p^n, v^n) . Finally, let j be the index of some commodity

with maximal bang-for-the-buck at (p^0, v^0) , and let b be that maximal bang-for-the-buck. Then when constructing an "near solution" to the EMP at (p^n, v^n) , you will look at x^0 , but with the amount of commodity j adjusted to just give utility level v^n . And you will look at x^n with the amount of j adjusted to just give utility level v^0 . Since $b > 0$ (which is the point of the assumption that some marginal utility is strictly positive), you know how large those adjustments must be (to the first-order), which gives you immediate bounds (to first-order terms) on the difference between $e(p^0, y^0)$ and $e(p^n, y^n)$. From here, it is a matter of cranking away.

■ 11.3. Use the identity $\nu(p, e(p, v)) = v$ and apply the Implicit Function Theorem. (Only the prices are variables here: v is treated as a constant.) The Implicit Function Theorem in this simple setting requires that $\partial\nu/\partial y$ is "nonsingular," which, in this context, means nonzero. The problem assumes that this partial derivative is strictly positive, so that's no problem.

■ 11.5. In Proposition 11.4, the utility function u is continuous and locally insatiable. In Proposition 11.9 (and, by incorporation, Proposition 11.8), I *do not* show that, if ν is an indirect utility function with all the properties listed, the function u that is defined by 11.6 is locally insatiable.

Since these propositions do establish that u is continuous, we will have local insatiability if we establish that u is strictly increasing for strict increases in its argument. (See Problem 10.5.) In fact, what I'll show is that, for $e \in R_+^k$ defined as $(1, 1, 1, \dots, 1)$, for all x , $u(x) < u(x + \delta e)$, for all $\delta > 0$.

I do this by contradiction. Suppose that, for some x and some $\delta > 0$, $u(x) \not< u(x + \delta e)$. Since u is shown to be nondecreasing, this means $u(x) = u(x + \delta e)$ and, in fact, $u(x') = u(x)$ for all x' in the "box" $x + \delta e \geq x' \geq x$. We know that $X_v = \{x \in R_+^k : u(x) \geq v\}$ is closed and convex (and comprehensive) for all v in the range of u , so setting $v^0 = u(x)$, this means we can put an open neighborhood around x that is disjoint from X_v for all $v > v^0$. Therefore, this open ball is disjoint from $\cup_{v > v^0} X_v$. Now $\cup_{v > v^0} X_v$ is a convex set: since the sets X_v nest, any two points z and z' from the union come from X_v for some $v > v^0$, but this X_v is convex, hence all convex combinations of them are in this X_v , hence in the union over all the X_v , $v > v^0$. The closure of this union is therefore convex (the closure of a convex set is convex) and is disjoint from x , so we can pass a strictly separating hyperplane between $\{x\}$ and the union. That is, for some $p \in R^k$, $p \cdot x < \inf_{z \in \cup_{v > v^0} X_v} p \cdot z$. By the usual argument (the union is comprehensive upwards) $p \geq 0$, and by perturbing p a bit, we can assume $p \in R_{++}^k$.

Now evaluate $\max\{u(z) : z \in R_+^k, p \cdot z \leq p \cdot x\} = \nu'(p, p \cdot x)$, where the notation ν' comes from Proposition 11.8. Proposition 11.8 tells us that this must be $\nu(p, p \cdot x)$. But we know from how p was constructed that this is also $u(x) = v^0$, since for all z that give higher utility than v^0 , $p \cdot z \geq$ the infimum $> p \cdot x$. And for ϵ small enough so that $p \cdot x + \epsilon$ is strictly less than the infimum, $\nu'(p, p \cdot x + \epsilon) = v^0$, for the same reason; $p \cdot x + \epsilon$ is, by design insufficient to afford any bundle with utility greater than v^0 . But

$\nu'(p, p \cdot x + \epsilon) = \nu(p, p \cdot x + \epsilon)$, and $\nu(p, p \cdot x + \epsilon) > \nu(p, p \cdot x)$, since ν is, by assumption, strictly increasing in its last argument. We have the desired contradiction, and hence the missing element of the proof.

■ 11.7. By inspection, this system of (alleged) Marshallian demand satisfies Walras' Law with equality and is homogeneous of degree zero in (p, y) . Moreover,

$$\frac{\partial x_1}{\partial p_1} + x_1 \frac{\partial x_1}{\partial y} = -\frac{y}{(p_1 + 2p_2)^2} + \left(\frac{y}{p_1 + 2p_2}\right) \left(\frac{1}{p_1 + 2p_2}\right) = 0,$$

$$\frac{\partial x_1}{\partial p_2} + x_2 \frac{\partial x_1}{\partial y} = -2\frac{y}{(p_1 + 2p_2)^2} + \left(\frac{2y}{p_1 + 2p_2}\right) \left(\frac{1}{p_1 + 2p_2}\right) = 0,$$

and, similarly, the other two terms of the Slutsky matrix are zero. The matrix of all zeros is clearly symmetric and negative semi-definite, so subject to the unstated technical conditions that give us the Integrability Theorem, these (alleged) Marshallian demand functions pass the test—there is a utility function that gives rise to them.

And, in fact, it isn't hard to see what "that" utility function must be (where "that" is in scare quotes because, of course, there are other utility functions that give rise to this demand—I'm going to give you the unique nondecreasing and quasi-concave utility function that does this.) The key to seeing what this utility function must be is that the Slutsky matrix is identically zero: If you shift a price and hold the consumer's utility fixed, she doesn't change her demand. Instead, she continues to demand goods 1 and 2 in a 1 to 2 ratio. This sort of demand goes with indifference curves that form right angles along the line $x_2 = 2x_1$, which in turn goes with the utility function $u(x_1, x_2) = \min\{x_1, x_2/2\}$.